

Alexander (Xander) Neary

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PROFILE

Software developer with experience building VR, simulation, and interactive systems using Unity, C#, Python, and OpenCV. Background in real-time graphics, procedural simulation, robotics control systems, and object-oriented software development. Recognized for balancing technical leadership, tutoring, and project development while managing a full academic workload.

EXPERTISE

- **Programming Languages:** C++, Python, C#, JavaScript
- **Frameworks & Tools:** Unity, OpenCV, Git, GitHub, Docker, Solidworks
- **Technologies:** Real-Time Simulation, Shader Programming, Object-Oriented Design, VR & Game Development
- **Practices:** Debugging, Agile/Scrum, Test-Driven Development

PROFESSIONAL EXPERIENCE

Software Developer – Summer Undergraduate Research Experience (SURE), USC Jun. 2024 – Aug. 2024

- Designed audiovisual systems in Unity (C#) for a VR simulator supporting chronic pain research studies.
- Engineered procedural wave simulation and real-time environment systems using custom shaders, improving visual realism and runtime stability.
- Developed and integrated Unity Editor automation tools, reducing simulation setup time by ~40%.

Team Manager – UCLA Esports: Overwatch Main Sep. 2025 – Present

- Managed operations and scheduling for a competitive collegiate team, coordinating across players, coaches, and league organizers.
- Introduced structured workflows and communication systems to improve preparation efficiency and team coordination.

Lead Student Instructor – UCLA Engineering 1: Bionics Sep. 2023 – Present

- Led weekly labs for 20+ students covering C++, Arduino, CAD, and hardware integration.
- Mentored and supervised a team of two to three instructors per term, providing guidance on classroom management.
- Diagnosed software-hardware integration issues, reinforcing object-oriented design and system-level debugging skills.

PROJECTS

3D Snake – Web Graphics Project 2026

JavaScript, CSS, WebGL, TinyGraphics/Three.js, GitHub – [Repository](#)

- Developed a 3D browser game featuring frame-independent physics and dual-perspective controls, optimized for real-time WebGL rendering.
- Engineered a responsive frontend UI/UX using CSS and JavaScript to manage game states, scoring, and interactive menus.

Robotic Hand-Tracking Control System 2025

Python, OpenCV, MuJoCo, GitHub – [Repository](#)

- Built real-time hand-tracking pipeline using OpenCV to control a simulated robotic manipulator.
- Integrated motion tracking with MuJoCo physics and a PID controller to enable stable, real-time robotic manipulation.

“On My Own” – 2D Platformer Game 2023

C#, Unity, itch.io, Visual Studio Code, GitHub – [Itch.io](#)

- Developed player movement systems and physics-based mechanics in Unity (C#).
- Designed localized gravity-field mechanics affecting traversal and gameplay dynamics in a collaborative team environment.

EDUCATION

University of California, Los Angeles (UCLA)

Sep. 2022 – Dec. 2026

Bachelor of Science in Computer Science

Los Angeles, CA

GPA: 3.2

- Relevant Coursework: Robotics, Computer Network Fundamentals, Computer Graphics, Computer Animation
- Organizations: Center for Excellence in Engineering and Diversity (CEED), UCLA Esports

INTERESTS

- Competitive esports, game development, beach volleyball, video editing